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Mesh Insensitive Structural Stress Method for Fatigue Analysis of Welded Joints Using the Finite Element Method

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Abstract

It is critical to determine initiation and early propagation of cracks for welded structures under fatigue loading. This is recognized by the local stress concentration at welded joints because of their domination to fatigue behavior of welded structures. Adopting the finite element (FE) method for determining the stresses for fatigue life calculations is well known approach. However, the result from FE analysis is highly sensitive to modelling techniques. The resulting stresses may differ depending on the size, type and integration order of elements. To overcome this dependency, Battelle has developed a mesh insensitive Structural Stress (SS) Method.

The method is based on the calculation of the stresses using the balanced nodal forces and moments at the weld toe location from the FE solutions. The resulting stresses will be regardless of size, type and integration order of elements. In this work, the essence of the SS method is studied. An existing problem of mesh sensitivity is considered to illustrate the application of the approach. The main purpose of this study is to perform SS method to this problem and to get mesh size independent results of stresses.

Keywords: Fatigue Life of Welded Structures, Structural Stress Method.

1. INTRODUCTION

Welding is the most common joining technique for metallic structures owing to its applicability to many geometric configurations. Fatigue cracking is the principal failure mode in welds. Therefore, it is critical to determine early propagation of cracks under fatigue loadings. Local stress concentration at welded joints dominate the fatigue behavior of welded structures. This stress concentration can be estimated through finite element (FE) method. However, the result from FE analysis is highly sensitive to modelling techniques. The resulting stresses may differ depending on the size, type and integration order of elements. To overcome this dependency, Battelle [1] has developed a mesh insensitive Structural Stress (SS) Method. In this work, after a brief review of some fatigue assessment methods, the essence of the SS method is studied with an existing problem of mesh sensitivity. The main purpose of this study is to perform SS method to this problem and to get mesh size independent results of stresses.

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2. FATIGUE ASSESSMENT METHODS OF WELDED JOINTS

A variety of fatigue assessment methods have been provided to estimate the fatigue life of welded joints under fatigue loading. The nominal stress approach is the simplest and most commonly applied method. Overall elastic behavior is assumed in this method. Nominal stresses are calculated in the sectional area under consideration and the local stress raising effects of the welds are disregarded. On the other hand, the stress raising effects of the geometrical configurations such as large cutouts are included [2]. When the geometry of the structure is complex, this method is not applicable [3].

The hot-spot stress approach has been developed to provide fatigue strength of welded structures where the nominal stress method is hard to apply because of geometrical complexities. The basis of the hot-spot stress approach is to base the fatigue assessment on the stress at the critical reference points that are called "hot spot" points defined by Niemi and Fricke [4, 5]. The use of the hot-spot stress method for the fatigue assessment of welded complex structures has become very prevalent with the increasing use of the finite element (FE) method. However, as the hot-spot stresses are often calculated in an area of high strain gradients, the result of FE analysis is highly mesh sensitive. The resulting stresses may differ depending on the size, type and integration order of elements.

Alternatively, the effective notch stress approach proposed by Radaj [6] has been used in fatigue assessments. The fatigue strength of welded joints is highly subjected to notch properties. Notches give higher stress concentrations which results with lower fatigue life. This method is mainly relied on the computed highest elastic stress at the critical points such as crack initiation points and weld toes. When using the effective notch stress method, meshed model with sufficient accuracy that can be obtained generally by 3D solid elements is required to capture the maximum stress at the point of stress concentration. All of these approaches can be seen in a stress distribution diagram shown in Fig. 1.

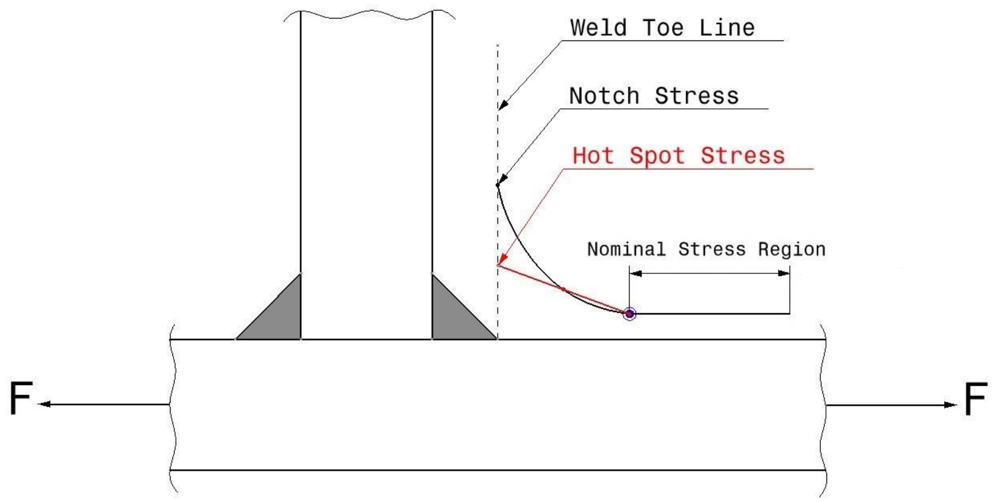


Fig. 1. Stress distribution for some fatigue assessment methods

3. STRUCTURAL STRESS METHOD

The Structural Stress Method is presented by Dong [7]. This approach is consistent with elementary structural mechanics theory. The method is based on the calculation of the stresses using the balanced nodal forces and moments at the weld toe location from the FE solutions. The resulting stresses will be regardless of size, type and integration order of elements.

The structural stresses are calculated in terms of membrane and bending components. First of all, a local coordinate system has to be defined, which is appropriate for calculating the structural stresses with respect to weld. According to this coordinate system nodal forces and moments are substituted in to:

$$\sigma_s = \sigma_m + \sigma_b = \frac{f_{x'}}{t} + \frac{6(m_{y'} + \delta f_{z'})}{t^2} \tag{1}$$

where σ_s is the structural stress, σ_m is the membrane component, σ_b is the bending component, t is the thickness of the element, $m_{y'}$ is the nodal moment with respect to defined local coordinate axis of y' and $f_{x'}$ and $f_{z'}$ are the nodal forces with respect to defined local coordinate in x' and z' direction, respectively. This parameters can be seen in Fig. 2.

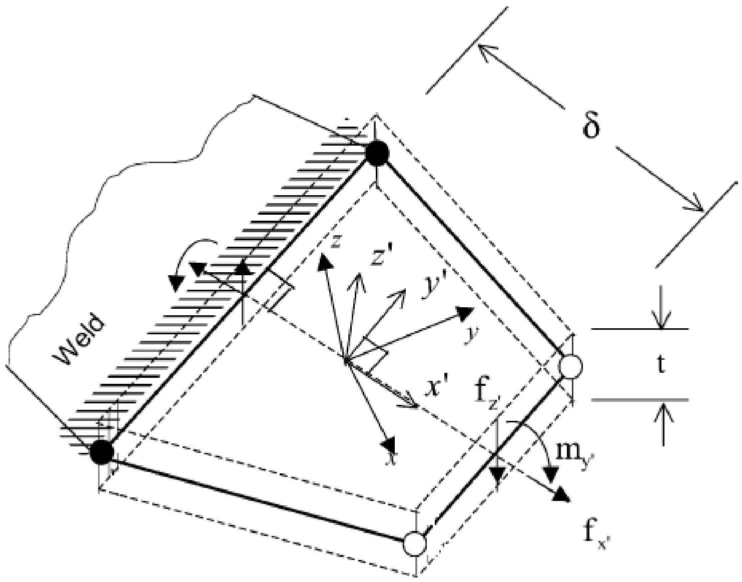


Fig. 2. Illustration of the structural stress method for an element [7].

4. PROBLEM DESCRIPTION

It is usual to use stiffeners in body of machines to reinforce structures in industry. For this reason benchmark geometry is chosen as shown in Fig. 3. In geometry, 200×200×600 mm square base profile are connected to 600×200 mm plate and for better strength 160×200 stiffener is welded as shown in Fig. 3. The thickness for all parts are 5 mm.

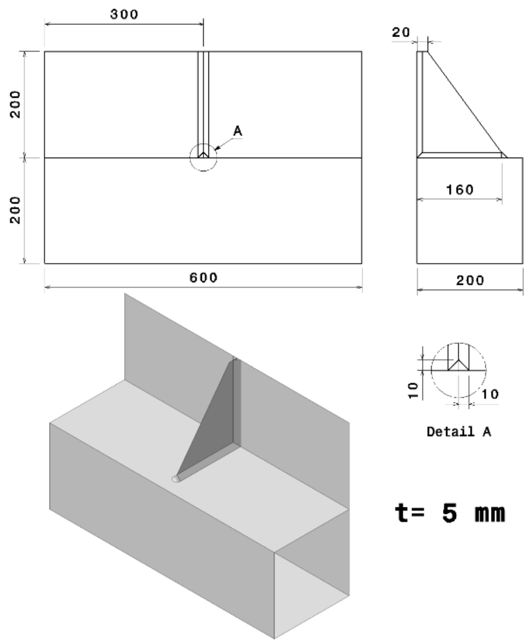
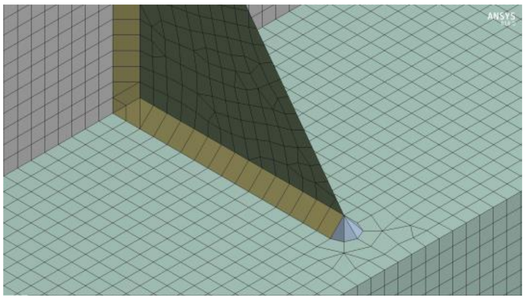
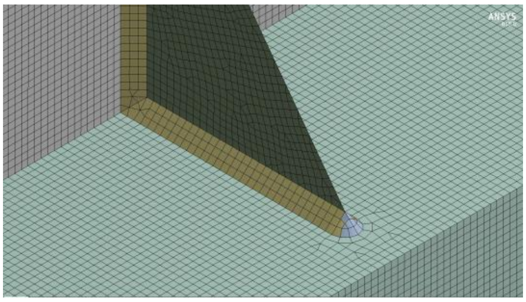


Fig. 3. Benchmark geometry detail

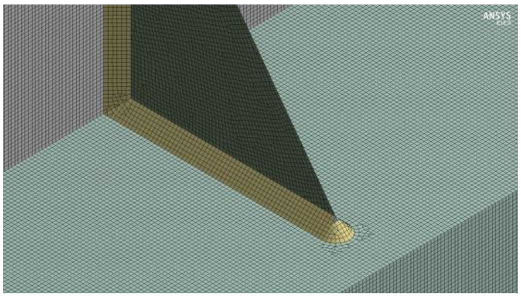
Then this geometry is meshed with four different mesh sizes. Models with different mesh sizes (10, 5 , 2.5 and 1.25 mm) are shown in Fig. 4. Considering the accuracy of the results, it is vital to use quad elements at the weld toes for all models. Shell 181 element type from ANSYS element library is used for shell elements. Number of the nodes and elements for overall finite element models of four different models is depicted in Table 1. Models are made from Structural Steel and the parameters for the material model are listed in Table 2.



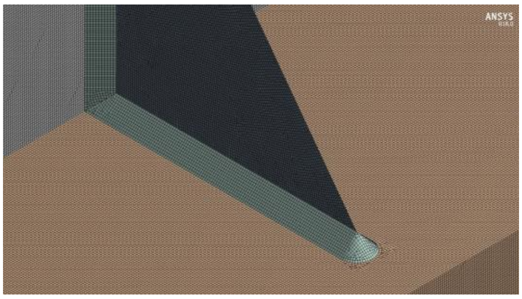
a- 10 mm



b- 5 mm



c- 2.5 mm



d- 1.25 mm

Fig. 4. Models with different mesh sizes

Table I. Finite element models detail.

Mesh size (mm)	Element type	Number of Nodes	Number of Elements
10	Linear Quad & Tria	6283	6280
5	Linear Quad & Tria	25243	25232
2.5	Linear Quad & Tria	100815	100797
1.25	Linear Quad & Tria	402485	402458

Table II. Material properties.

Properties	Value	Units
Density	7850	kg/m^3
Young's Modulus	200000	MPa
Poisson's Ratio	0.3	-
Yield Strength	250	MPa
Ultimate Strength	460	MPa

5. RESULTS AND DISCUSSION

The finite element analysis of this study is conducted using ANSYS Mechanical V.18. The loading and boundary conditions are defined as shown in Fig. 5. The model is fixed from the end of 200×200 square base profile and 5000 N force are applied to 200×600 plate in –y direction.

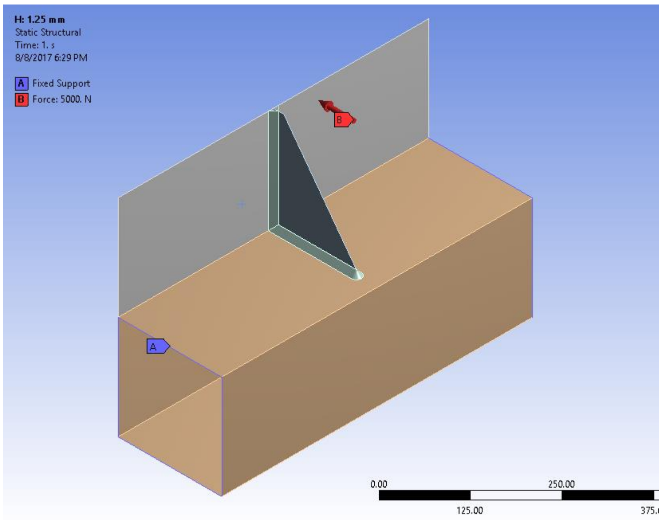


Fig. 5. Loading and boundary condition of model

Under these conditions, problems are solved with finite element solver. Fig. 6 shows the von Mises stress results of the models. Elements with the highest v.Mises stresses are pointed with an arrow.

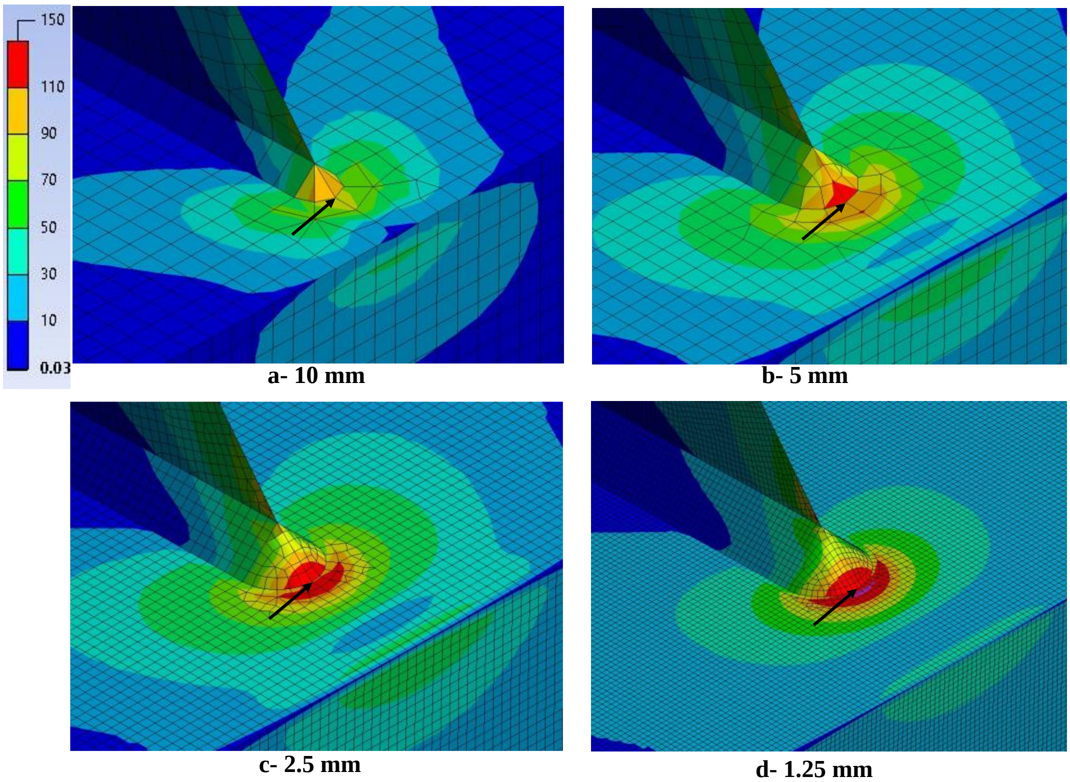


Fig. 6. FE Stress (von Mises) results

Mechanical APDL Module of ANSYS V.18 is used for reading the nodal forces and the nodal moments. Using Eq. (1), Structural Stresses are calculated for each model separately. The results are depicted in Table 3. Variation in the Structural Stresses with different element sizes is 0.88% whereas von Mises stresses at the critical points with same mesh sizes have quite large range of stress. Besides, it can be seen from the table that with decreasing the element size –improving the mesh quality- von Mises stresses are increasing considerably. On the other hand, SS results are nearly same at higher stress values from the von Mises stresses.

Table III. Results

Element size	Stress von misses (Mpa)	Variation (%)*	Structural Stress (Mpa)	Variation (%)*
10	107.14	-16.22	151.39	-0.0726
5	129.30	1.11	151.36	-0.0924
2.5	135.72	6.13	150.50	-0.6600
1.25	139.35	8.96	152.73	0.8118
Average Stress(Mpa)	127.88	**25.19** **	151.5	**0.8844** **

* Variation(%) =[(Stress - Average Stress) / Average Stress] ×100

** Total Variation = Max Va. – Min Va.

6. CONCLUSION

This study considers an existing problem of mesh sensitivity to illustrate the application of the Structural Stress approach. A workbench geometry is meshed with four different element sizes and finite element analysis of these models were successfully conducted. Based on the results obtained in this study, it is proved that SS method is mesh insensitive and it can be used for an alternative fatigue assessment method.

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